# Lian Yang

**Full-Stack & AI Automation Engineer**

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## SUMMARY

Informatics Engineer with ~2 years of hands-on full-stack development in React and PostgreSQL. Builds end-to-end AI pipelines (LLMs, speech recognition, TTS, image generation), workflow automations with n8n, and containerized services with Docker and Kubernetes. Ships tested, documented, security-conscious software; focused on remote full-stack, backend and AI-automation roles.

## TECHNICAL SKILLS

**Frontend:** React, JavaScript, HTML/CSS

**Backend & Data:** Python (FastAPI, Typer, Pydantic), PostgreSQL, REST APIs

**AI & Automation:** LLM APIs (OpenAI-compatible), structured JSON outputs, Whisper ASR, TTS, image generation, AI workflows, n8n

**DevOps & Infra:** Docker, Kubernetes, server deployment & maintenance, CI/data pipelines, load balancing, distributed systems (intermediate)

**Security & Quality:** Secret scanning, input validation, path-traversal protection, pytest, headless test suites

**Other:** Algorithms, FFmpeg, Godot/GDScript, Git/GitHub

## PROJECTS

**Codex Stickman Pipeline — open source (MIT) · github.com/CatbearLang/codex-stickman-pipeline**

- Designed and built a Python CLI that turns a topic into a narrated 1080p explainer video in one command: LLM script → TTS → word alignment → AI images → FFmpeg render → automated verification.
- Implemented fail-closed cost controls (explicit --live flag, durable budget ledger, pre-call image caps) and resumable jobs with OS locks and a job journal.
- Delivered a fully offline test suite (94 passing tests) with provider test doubles, plus a release secret scanner; documented architecture in English and Spanish.

**GlosSub — bilingual subtitle generator (Python, FastAPI, Whisper, LLM)**

- Architected a 6-stage cached pipeline (extract, transcribe, segment, translate, align, render) producing color-aligned bilingual .ass/.srt subtitles.
- Achieved 99% translated cues and 89% phrase-aligned coverage (688 groups) on a 23.7-min episode; cut alignment time to ~11 min through batching and concurrency tuning; cached re-renders run in under 1 s.
- Secured the local web UI: loopback-only binding, fail-closed path-traversal checks, secrets only from environment, LLM output validated with Pydantic.

**Arcane Knockout — 3D multiplayer arena game (Godot 4.7, GDScript)**

- Built the game logic and code with an AI-assisted workflow (art and 3D models AI-generated): rules, physics-based knockback system, AI bots, in-game economy and online multiplayer (ENet/UDP, room codes, LAN discovery, UPnP).
- Maintained ~190 GDScript files with a headless automated test runner (e.g. 519 checks and 349 economy checks, 0 failures) required before every Android build; shipped Windows and Android releases.

**n8n automation pilots**

- Designed and tested workflow automations in n8n integrating APIs and AI steps.

## PROFESSIONAL EXPERIENCE

**Junior Programmer · 3 years**

- Developed and maintained web application features across frontend (React) and backend, working with PostgreSQL databases.
- Performed system maintenance, debugging and support for existing software, collaborating on fixes and improvements.

**IT Support — Vinader Computación, Encarnación**

- Provided hardware and software support: installation, configuration and troubleshooting of computers and peripherals.
- Performed preventive and corrective maintenance of equipment and systems.

**Cashier — Casa Jorge, Encarnación · 5 years**

- Provided customer service in a retail store.
- Handled cash, payments and daily register balancing with accuracy and responsibility.

## EDUCATION

- Informatics Engineering — Universidad Nacional de Itapúa, Encarnación
- Technical High School Diploma in Informatics (Bachiller Técnico en Informática) — Colegio Experimental de la Universidad Católica, Encarnación, 2015–2017 · Distinguished Student honors, 2016 and 2017
- English Intermediate 3A — EUROSUR Training Center (100 hours), 2015

Languages: Spanish (native) · English (intermediate)